

Database Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It involves dividing large tables into smaller, related tables and defining relationships between them using foreign keys.

The main goals of normalization are to eliminate redundant data, ensure data dependencies make sense, and reduce the chances of data anomalies during insert, update, and delete operations.

First Normal Form (1NF)

A table is in First Normal Form if it contains only atomic (indivisible) values, and each record is unique. This means no repeating groups or arrays are allowed within a single column.

Example: A table storing a student's phone numbers should not have a single column containing multiple comma-separated numbers. Instead, each phone number should be in its own row or a separate related table.

Second Normal Form (2NF)

A table is in Second Normal Form if it is already in 1NF and all non-key attributes are fully functionally dependent on the entire primary key, not just part of it. This rule mainly applies to tables with composite primary keys.

Example: In an Orders table with a composite key of OrderID and ProductID, if ProductName depends only on ProductID (not the full composite key), this is a partial dependency and violates 2NF. ProductName should be moved to a separate Products table.

Third Normal Form (3NF)

A table is in Third Normal Form if it is in 2NF and has no transitive dependencies, meaning non-key attributes do not depend on other non-key attributes.

Example: In a Students table, if StudentID determines DepartmentID, and DepartmentID determines DepartmentName, then DepartmentName is transitively dependent on StudentID. This should be split into a separate Departments table.

Boyce-Codd Normal Form (BCNF)

BCNF is a stricter version of 3NF. A table is in BCNF if, for every functional dependency X to Y, X must be a super key. This handles certain edge cases that 3NF does not fully address, particularly when a table has multiple overlapping candidate keys.

Sample Practice Question

Q: Given a table Enrollment(StudentID, CourseID, StudentName, CourseName, InstructorName) where InstructorName depends only on CourseID, identify the normalization violation and suggest a fix.

Answer: This violates 2NF because InstructorName depends only on part of the composite key (CourseID), not the whole key (StudentID, CourseID). Fix: Split into Enrollment(StudentID, CourseID), Students(StudentID, StudentName), and Courses(CourseID, CourseName, InstructorName).